

Education

University of Utah*B.S. Computer Science***Salt Lake City, UT***August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++ , SQL, Java

Software: React, Node.js, AWS, SageMaker, Fargate, Docker, Podman, REST, Django, PyTorch, MongoDB, Linux

Experience

L3Harris Technologies*Associate Software Engineer***Greenville, TX***June 2025 – Present*

- Built containerized microservices for high-frequency data ingestion, processing, and storage using Podman and serverless architecture on defense sensor streams.
- Developed signal processing algorithms in Python and C++ on embedded systems, meeting strict latency and resource constraints in collaboration with hardware teams.
- Automated CI/CD pipelines with static code analysis and integration testing, reducing deployment friction across engineering teams.

Doxy.me*Software Engineering Intern***Charleston, SC***August 2024 – May 2025*

- Built transformer-based models with PyTorch and Llama, integrating WebRTC for real-time video streaming.
- Deployed inference pipelines on SageMaker & Fargate, improving diagnostic accuracy with reinforcement learning.
- Authored and maintained YAML-based configuration files for CI/CD pipelines, cloud infrastructure, and application deployments, enabling reproducible environments and streamlined releases.

University of Utah*Undergraduate Research Assistant – FuTURES Lab***Salt Lake City, UT***May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

HEXstream*Software Engineering Intern***Chicago, IL***May 2022 – Aug 2022*

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
 - Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.
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Projects

University Course Planning/Review Platform: Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend using React + Tailwind, including interactive dashboards, heatmaps, calendar visualizations, and search interfaces.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

Ray Tracing Engine: Created an interactive WebGL-based ray tracing engine in JavaScript, featuring realistic reflections, dynamic lighting, and customizable environment maps. Implemented shaders and user controls for rendering techniques, scene adjustments, and interactivity.